

DATA612 Decision Analytics
Group Assignment #2 – Konys, Inc.
In-Class Session: March 25, 2026
Report Submission: March 30, 2026

Answer the following questions based on the Konys, Inc. case study. In addition to the information given in the case, assume that there are fixed operating expenses accruing in each quarter. Selling, general, and administrative expenses are \$37,920,000. Design and research expenses are \$17,837,500 per quarter.

Note that this setting is slightly different than the inventory management environment in the Halloween example that we discussed in class. In that example, any unmet demand was a lost sale. In Konys, Inc. case, there is no lost sales. Any unmet demand is satisfied either from the spot market or the reserved option, depending on the contract type and the prices transpired. Note that the option is exercised only if it is available and if it is cheaper than the spot market price.

Part A. (50 Points)

A1. Develop a Monte Carlo Simulation Model for a single quarter of a long-term purchase contract that is explained in “Part A” of the case study. Your model could be an Excel spreadsheet or a Python function that calculates the total profit given the order size, the realized demand, and the realized spot price. Next, you will be asked to implement this model for Q3.

A2. Use the Python Fitter library to check the probability distributions that could be used to model the historical Q3 sales values and the spot prices. Retrieve the parameters of a Normal distribution that fits to the Q3 sales data and the parameters of a Beta distribution that fits to the spot price data. Use these distributions in your simulations in the questions below whenever necessary.

A3. Estimate the total expected profit of the Q3 purchase contract by simulating your model if they ordered 2,170,000 units. Use at least 10,000 iterations.

A4. Find an optimal ordering quantity with your model by exploring alternative order quantities. What is the optimal total expected profit? Develop a 90% confidence interval for this value. You can explore with the increments of 100,000 units.

Part B. (50 Points)

B1. Develop a Monte Carlo Simulation Model for a single quarter of an option contract that is explained in “Part B” of the case study. Your model could be an Excel spreadsheet or a Python function that calculates the total profit given the order size as well as the realized demand and a spot price. Next, you will be asked to implement this model for Q3.

B2. What should Konys set the option contract quantity for Q3 if they place an order of 2,170,000 units? What is the optimal total expected profit? Develop a 90% confidence interval for this value. You can explore with the increments of 100,000 units.

B3. Compare the solutions you found in A3 and B2 from total profit and risk perspectives. For risk performance, use percentiles.

B4. What should Konys set the option contract quantity for Q3, if their order quantity was what you suggested in A4? What is the optimal total expected profit? Compare this solution with the solution of A4. Which one yields more profit? Why?

Guidelines and Rubric:

1. I expect that Part A and Questions B1 of this assignment are completed in-class performance, which add up to 70% of this assignment.
2. Teams should continue to work on Part B in-class as time permits, but the grading for these parts will mainly be based on the report that will be submitted later.
3. Each team will present their progress by the end of the class starting at 7:15 pm. Each team will have four minutes to present. Presentations will take place between 7:15 pm – 7:45 pm. Each team will submit their models and algorithms at 7:15 pm.
4. Your report should include all the answers to the questions asked above. The report should be in a business format that can be submitted to the executive board of Konys, Inc.
5. I can deduct up to 10% for poor style and formatting of the report.
6. All models and algorithms developed should be submitted along with the report.
7. It is academic misconduct to communicate with anyone else other than your own group members and the instructor about the assignment. Use of generative AI tools are also not permitted in solving the problem and developing the algorithms.